

Wiring & Interconnect Document

Unit Test / Integration Set — CU1, CU2, CU3 | Reworked, corrected & redrawn from handwritten source

Schematic convention. Relays are drawn as dashed relay blocks containing ganged normally-open (N.O.) contacts linked to a coil (terminals A1/A2). Filled dots are terminals; **[VERIFY]** marks readings that were unclear in the scan and need confirmation against the original.

Corrections applied from the source review: 1) "loom" was mis-typed as "100m" — no lengths exist, looms only. 2) Invented "Arrow/Standard loom" types removed. 3) RTD terminals shown as T1/T2/T3 (not $\pm 1/\pm 2/\pm 3$). 4) Fan Motor (M) added to GCF. 5) Unclear items flagged rather than guessed.

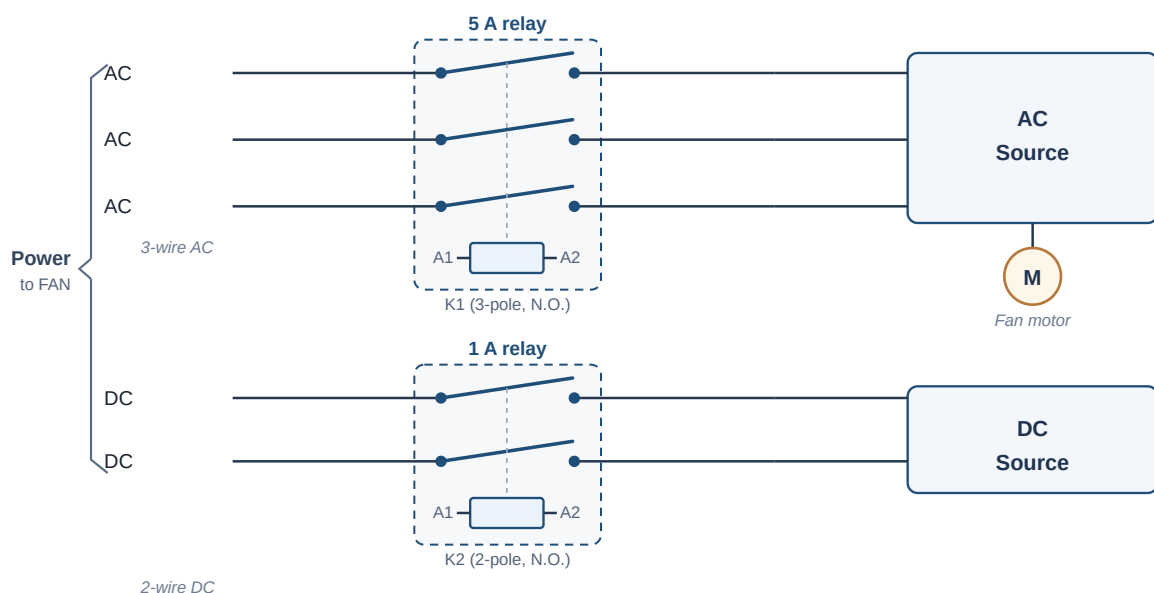
1. GCF — Fan Control Unit (CU3-P1)

1.1 Controller signals → DMM



- **Controller (CU3-P1)** — the unit under test; presents the fan FAN+ line and a FAULT line for measurement.
- **DMM** — digital multimeter; FAN+ and FAULT land on terminals A3+ / A3- for current/fault sensing.

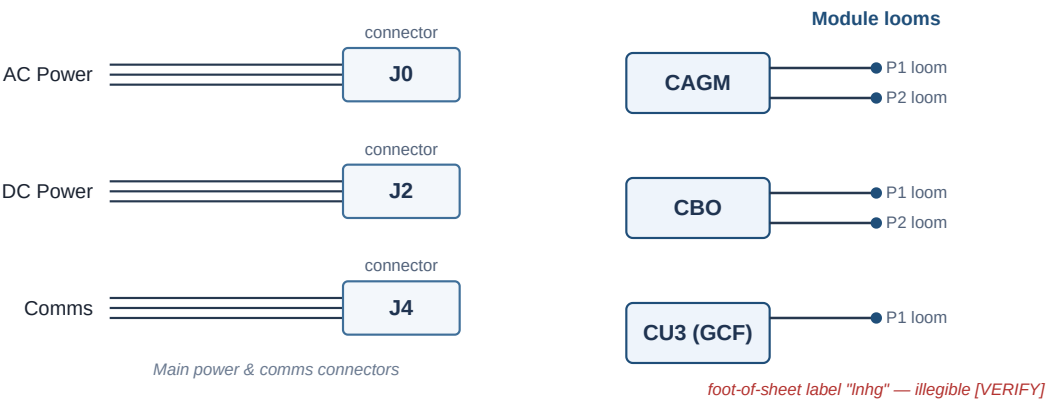
1.2 Switched power to the fan



- **5 A relay (K1)** — 3-pole N.O. contactor switching the 3-wire AC feed from the AC Source to the fan.
- **Fan motor (M)** — the AC load — the cooling fan driven through K1.
- **1 A relay (K2)** — 2-pole N.O. relay switching the 2-wire DC feed from the DC Source.
- **AC / DC Source** — programmable bench supplies feeding the unit.

From (signal)	Via	To	Notes
Controller: FAN+	direct	DMM A3+	current sense (+)
Controller: FAULT	direct	DMM A3-	current sense (-); "!" mark in source
Power to FAN: AC (3 wires)	5 A relay (K1)	AC Source → fan motor (M)	3-phase / 3-wire AC
Power to FAN: DC (2 wires)	1 A relay (K2)	DC Source	DC +/- feed

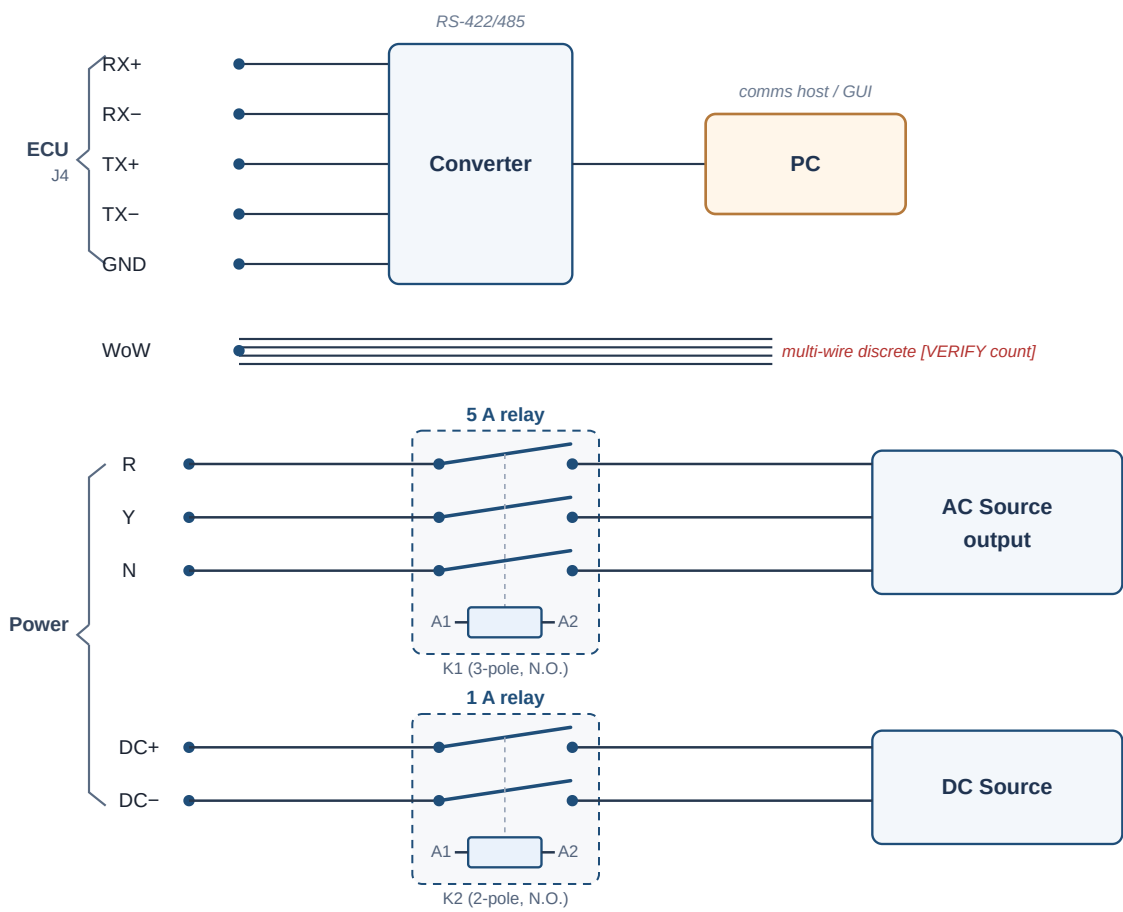
2. Integrated — Connectors & Loom Overview



- **J0 / J2 / J4** — main system connectors: J0 = AC power, J2 = DC power, J4 = comms (J4 matches the ECU comms connector on the CCM sheet).
- **Module looms** — wiring looms running to each module: CAGM (P1, P2), CBO (P1, P2) and CU3/GCF (P1).
- **CBO** — a further module referenced here only; no individual wiring sheet was supplied.

Signal / Module	Connector / Port	Loom	Notes
AC Power	J0	—	main connector
DC Power	J2	—	main connector
Comms	J4	—	matches ECU J4 (CCM)
CAGM	P1 / P2	loom	two looms
CBO	P1 / P2	loom	no individual sheet
CU3 (GCF)	P1	—	
"Inhg" (illegible)	—	—	[VERIFY] foot-of-sheet label

3. CCM — Communications Module (CU2-P2)

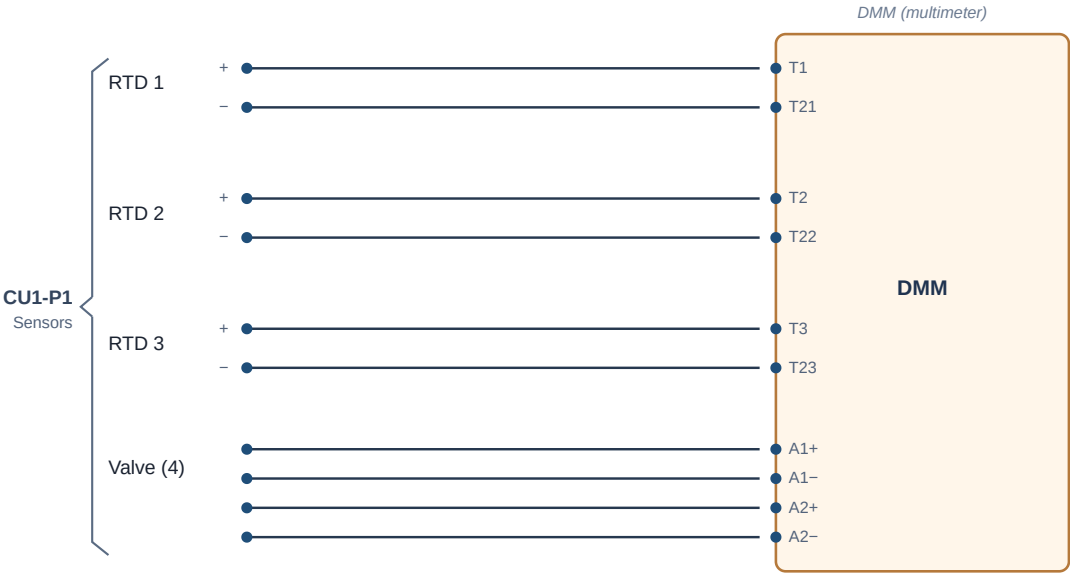


- **ECU J4** — serial connector exposing RX+, RX-, TX+, TX- and GND.
- **Converter** — RS-422/485-style serial converter between the ECU link and the PC.
- **PC** — comms host running the operator GUI.
- **WoW** — multi-wire discrete signal (count to confirm).
- **5 A relay (K1) / 1 A relay (K2)** — switch the AC (R/Y/N) and DC supplies from the sources.

From (signal)	Via	To	Notes
ECU J4: RX+ / RX-	Converter	PC	differential serial
ECU J4: TX+ / TX-	Converter	PC	differential serial
ECU J4: GND	Converter	PC	signal ground
WoW signal	direct (multi-wire)	—	[VERIFY] wire count
AC: R / Y / N	5 A relay (K1)	AC Source out	[VERIFY] possible B phase
DC+ / DC-	1 A relay (K2)	DC Source	[VERIFY] destination

4. CAGM — Sensor & Power Module (CU1)

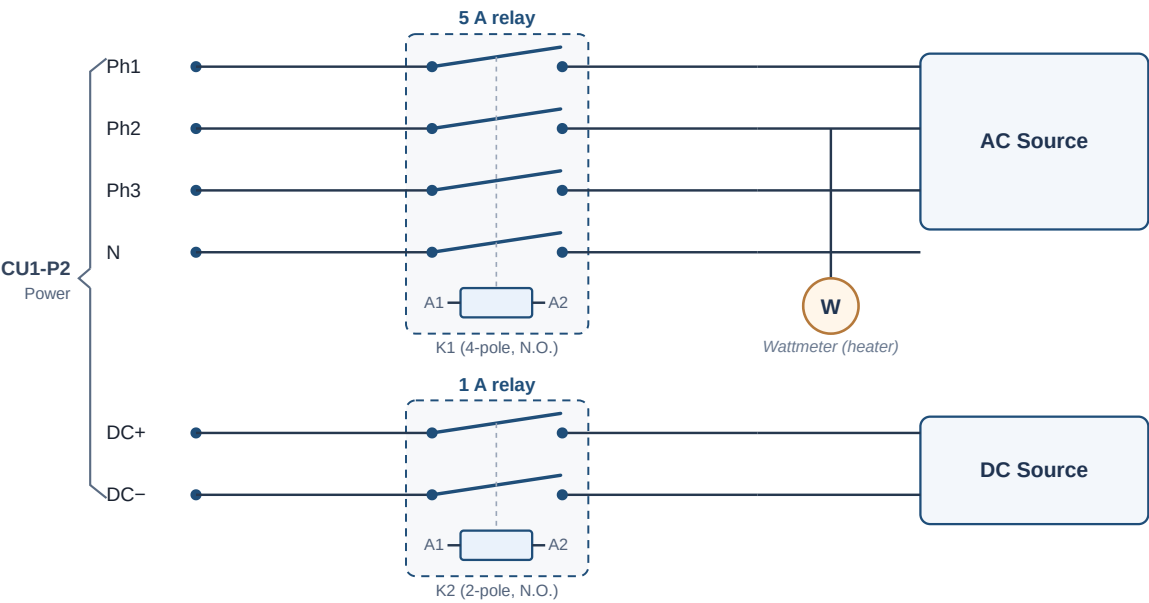
4.1 CU1-P1 — sensors → DMM



- **RTD 1–3** — three resistance temperature detectors; each + / – pair lands on DMM terminals T1/T21, T2/T22, T3/T23.
- **Valve (4)** — 4-wire valve monitored on A1+, A1–, A2+, A2–.
- **DMM** — measures all sensor channels.

Sensor	Lead	DMM terminal	Notes
RTD 1	+ / –	T1 / T21	[VERIFY] was ±1
RTD 2	+ / –	T2 / T22	[VERIFY] was ±2
RTD 3	+ / –	T3 / T23	[VERIFY] was ±3
Valve (4)	A1+ A1– A2+ A2–	DMM	4-wire valve control

4.2 CU1-P2 — switched power



- **5 A relay (K1)** — 4-pole N.O. contactor switching Ph1/Ph2/Ph3/N from the AC Source.
- **Wattmeter (W)** — measures power delivered to the heater load.
- **1 A relay (K2)** — switches the DC+/DC- feed from the DC Source.

Line	Via	To	Notes
Ph1 / Ph2 / Ph3	5 A relay (K1)	AC Source	load via Wattmeter (W)
N (neutral)	5 A relay (K1)	AC Source	
DC+ / DC-	1 A relay (K2)	DC Source	

4.3 GUI signal mapping

- **Temp. sensor** → **GUI** — read / display (monitoring).
- **Heater (W)** → **GUI** — read / display (monitoring).
- **Valve** → **GUI** — ON / OFF command (control action).

5. Items to Verify Against the Original

- RTD terminals: **T1/T2/T3** vs $\pm 1/\pm 2/\pm 3$ (§4.1).
- CCM AC phases: **R/Y/N** vs R/Y/B/N (§3).
- CCM WoW conductor count (§3).
- CCM DC return destination — DC Source vs AC Source (§3).
- Integrated foot-of-sheet label “Inhg” (§2).
- Expansion of acronyms GCF / CCM / CAGM / CBO and the “All/AU” prefix on GCF.

Relay references K1/K2 and pole counts are drawing aids inferred from the number of switched conductors; confirm against the parts list. Acronym meanings are not defined in the source and are described from the wiring only.